

module-07-eleven-multiplication - Darshana (60-min workshop)

IKS Curriculum - middle band

Module 7 — $\times 11$ Multiplication (Darśana)

Sanskrit: *Ekādhikena Pūrveṇa* · **Band:** Middle (6-8) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Mental multiplication by 11 from 20th-century Indian arithmetic.

Darśana note. This workshop condenses Module 7's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Show 87×11 mentally. Most students reach for long multiplication. “Or — *split, add, with carry*. Today's lesson: $\times 11$ mental method, with the carry case included.”

Core idea · 15 min

$\times 11$ method:

For $(10a + b) \times 11 = 100a + 10(a+b) + b$.

Procedure: 1. Split the 2 digits: $a _ b$ 2. Add them: $a+b$ 3. Put the sum in the middle: $a (a+b) b$

Carry case: if $a+b \geq 10$, carry the tens digit to the first slot. - $87 \times 11 = 8_ (8+7) _ 7 = 8_ 15 _ 7 \rightarrow$ carry: $(8+1) 5 7 = 957$.

Extension: 3-digit $\times 11$: $abc \rightarrow a (a+b) (b+c) c$.

Demonstration · 10 min

Live speed race: every student solves 5 problems mentally. Time it. Average ~3 seconds per problem with practice. Compare with the long-multiplication time for the same problems.

Source moment · 5 min

Paraphrase Tirthaji 1965: “*Ekādhikena Pūrveṇa* — ‘by one more than the previous.’ The *sūtra*-name for the $\times 11$ (and related) methods.”

Hands-on activity · 15 min

Mental-math sprint. Each student does 20 $\times 11$ problems in 3 minutes. Best score wins.

Reflection + take-home · 10 min

“When is $\times 11$ mental method faster than the long method? When is it slower? Why?”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — The $\times 11$ hook — what is the trick?
- **Day 02** — The carry case
- **Day 03** — Why it works — the algebra
- **Day 04** — 3-digit $\times 11$
- **Day 05** — Speed day
- **Day 06** — Anurūpyena extension to $\times 12$ -19
- **Day 07** — Limits of the method
- **Day 08** — Tirthaji 1965 and the historical claim
- **Day 09** — Mixed practice
- **Day 10** — Final assessment

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.